

EGCO 425 – Chapter 13

Association Analysis

References

- [1] Tan et al., Introduction to Data Mining (Chapter 6)
- [2] Han & Kamber, Data Mining: Concepts and Techniques (Chapter 6)
- [3] RapidMiner Doc, <https://docs.rapidminer.com/latest/studio/operators/>

What is association analysis ?

⊕ Given a set of transaction, find rules to predict the occurrence of an item based on the occurrence of the others

⊕ Examples of association rules

▶ {beer, bread} → milk

We see that a customer has bought beer & bread; it is likely that she will also buy milk

▶ {diaper} → beer

TID	Items
1	bread, coke, milk
2	beer, bread
3	bread, coke, diaper
...	...

⊕ We can deduce from the rules, for example,

- ▶ Which products help boost the sales of milk ? Which products should be promoted together with milk ?
- ▶ Which product will be affected if the supermarket stops selling bread ?

Outline

⊕ Association rule discovery

- ▶ Fundamental elements → itemset, support, confidence
- ▶ Frequent Itemset generation → Apriori, FP-growth
- ▶ Rule generation

⊕ Evaluation & interpretation

- ▶ Objective vs. subjective evaluation
- ▶ Common metrics, properties, precautions

Definitions

- ⊕ $I = \{I_1, I_2, \dots, I_m\}$ is an itemset or a set of items
 - ▶ k-itemset is an itemset containing k items
 - ▶ E.g. { bread, coke, milk } is a 3-itemset
- ⊕ $D = \{T \mid T \subseteq I\}$ is a set of transactions in a database
 - ▶ Transaction T contains an itemset A iff $A \subseteq T$
- ⊕ $A \rightarrow B$ or "if A, then B" is an association rule where
 - ▶ $A \subset I, B \subset I, A \cap B = \emptyset$
 - ▶ A = antecedent or premise, B = consequence or conclusion
 - ▶ Note that association analysis is a descriptive data mining task. It is not to predict conclusion (B) in the same way as predicting target class in classification. The association rule tells us that **if A occurs, B is also likely to occur (based on evidence in data set)**

Input Format

Market-Basket (Transaction) Format	
TID	Items bought
1	bread, coke, milk
2	beer, bread
3	bread, coke, diaper
4	...

Fixed-Length Record Format					
TID	Beer	Bread	Coke	Diaper	Milk
1	0	1	1	0	1
2	1	1	0	0	0
3	0	1	1	1	0
4	...				

- ⌘ In fixed-length format, attributes are treated as **asymmetric attribute**
- ⌘ There are so many items in supermarket. We usually focus on items being bought by customers, i.e. only when attribute values = 1 (true)

Voter	Issue 1	Issue 2	Issue 3	Issue 4	Issue 5
1	0	1	1	0	1
2	1	1	0	0	0
3	0	1	1	1	0
4	...				

0 = disagree
1 = agree

- ⌘ In fixed-length format, attributes are treated as **symmetric attribute**
- ⌘ Number of issues are usually small. We are interested in issues being both agreed and disagreed by voters, i.e. when attribute values = 0 (false) and 1 (true)

Support & Confidence

⊕ Support or coverage

- ▶ Fraction of transactions containing itemset(s), for example
 - $\text{Support}(\{\text{beer}\}) = 1/3$
 - $\text{Support}(\{\text{bread}\} \cup \{\text{coke}\}) = 2/3$
- ▶ In association analysis, $A \cup B$ means A and B co-exist. it is equivalent to $A \cap B$ in math

⊕ Given an association rule $A \rightarrow B$

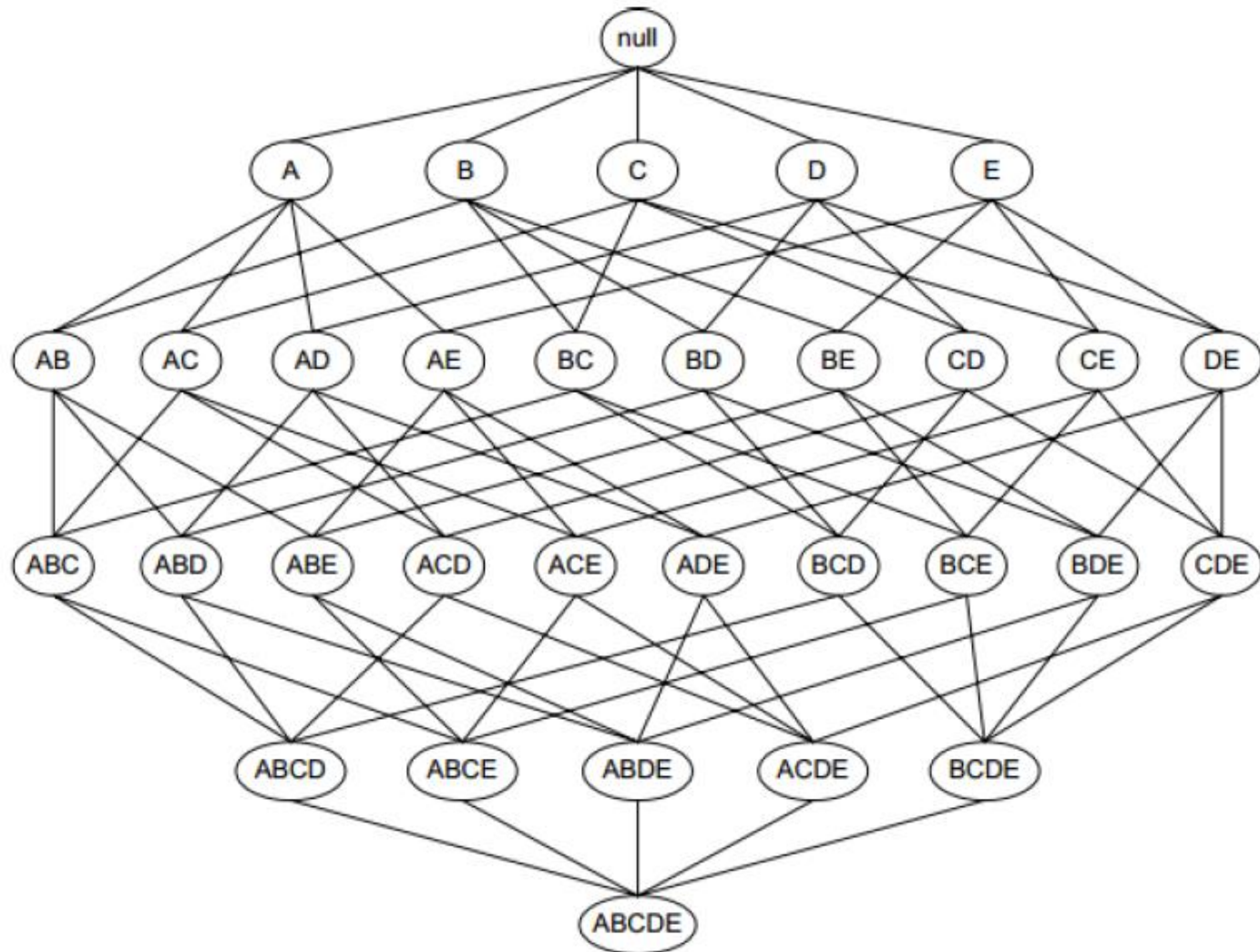
- ▶ $\text{Support}(A \rightarrow B) = \text{support}(A \cup B) = P(A \cap B)$
- ▶ $\text{Confidence}(A \rightarrow B) = \text{support}(A \cup B) / \text{support}(A) = P(B | A)$
 - = prob. that item B appears in transactions containing item A
 - = prob. that B will occur given that A has occurred

Finding Association Rules

- ⊕ Goal of association analysis is to find strong association rules
 - ▶ $\text{support}(A \rightarrow B) \geq \text{minSupport}$
 - ▶ $\text{confidence}(A \rightarrow B) \geq \text{minConfidence}$
 - ▶ minSupport and minConfidence are parameters given by users
- ⊕ Rationale for strong association rules
 - ▶ Rules with too low support may occur by chance & are likely to be uninteresting from a business point of view
 - ▶ Confidence measures the reliability of the inference made by the rule
- ⊕ Typical processing
 - ▶ Phase 1 : find all frequent itemsets that pass minSupport criterion
 - ▶ Phase 2 : generate rules by permuting frequent itemsets & keep rules that pass minConfidence criterion

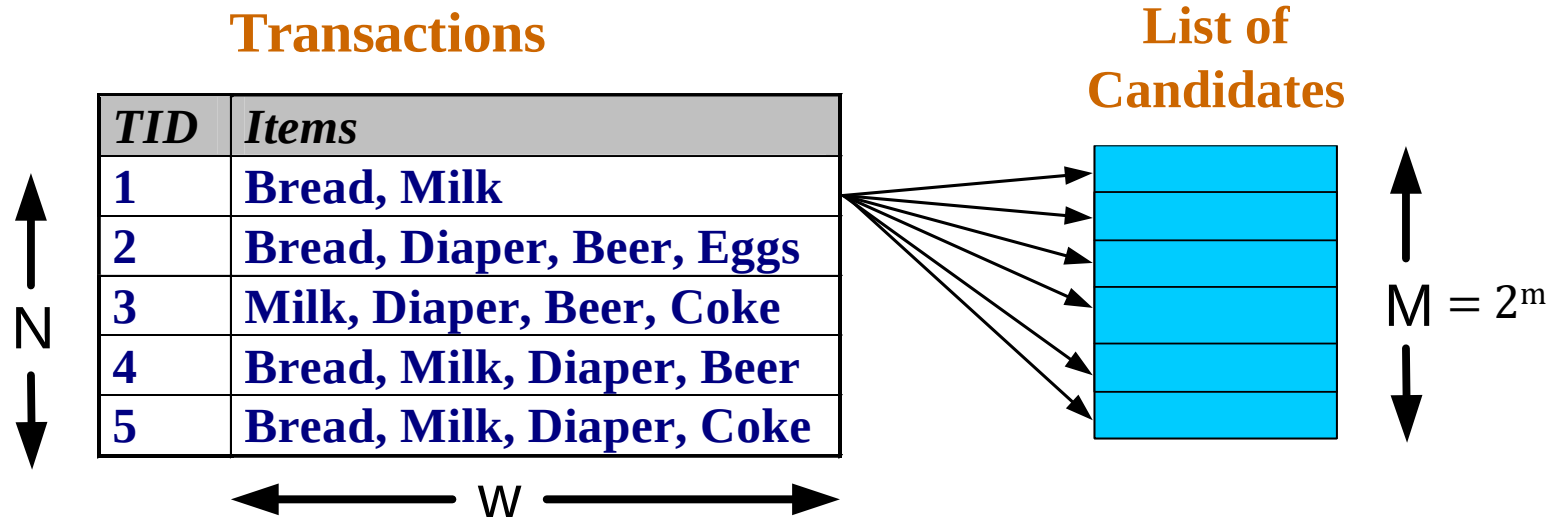
Phase 1 : Finding Frequent Itemsets

- ⊕ Brute force method is computationally expensive
 - ▶ First, we have to find all candidate itemsets
 - ▶ Then, for each candidate
 - Scan DB & count transactions containing all items in the itemset
 - If the count $\geq$ minSupport → keep it as frequent itemset
- ⊕ Alternative methods
 - ▶ Apriori
 - ▶ FP-growth



Given m items, there are 2^m candidates to consider

Brute Force Method



- ⊕ For supermarket with m products, N transactions, customers buy at most w products at a time (i.e. max transaction length)
 - ▶ Scan DB for 1 candidate $\rightarrow O(Nw)$
 - ▶ For all candidates $\rightarrow O(MNw) = O(2^m Nw)$

Monotone vs. Anti-Monotone Properties

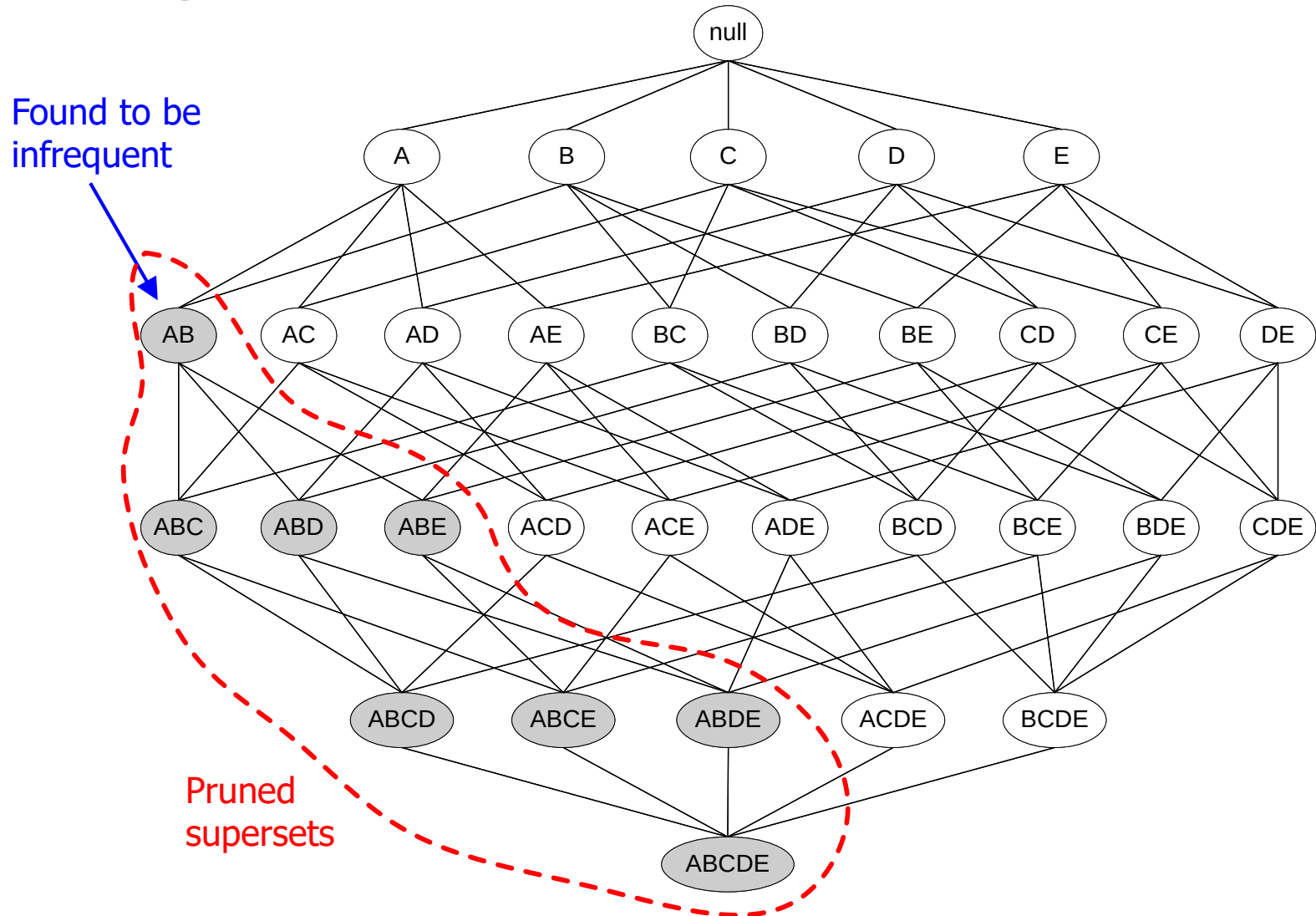
⊕ Monotone or upward-closed property

- ▶ If $(X \subseteq Y)$, then $F(X) \leq F(Y)$ i.e. smaller set, smaller function's value

⊕ Anti-Monotone or downward-closed property

- ▶ If $(X \subseteq Y)$, then $F(X) \geq F(Y)$ i.e. smaller set, bigger function's value
- ▶ Support(itemset) has anti-monotone property. For example
 - Few customers buy {Coke} → even fewer buy {Coke, Bread}
 - Many customers buy {Bread, Milk} → many buy {Bread} and many buy {Milk}

- ⊕ Apriori uses anti-monotone property of support to prune out some candidate itemsets. That is, if an itemset is not frequent, then its superset is also not frequent



Example : Apriori and support-based pruning

TID	Items bought
1	A B E
2	B D
3	B C
4	A B D
5	A C
6	B C
7	A C
8	A B C E
9	A B C

minSupport count ≥ 2

1-itemsets	
Itemset	Count
A	6
B	7
C	6
D	2
E	2

All pass minSupport

2-itemsets	
Itemset	Count
AB	4
AC	4
AD	1
AE	2
BC	4
BD	2
BE	2
CD	0
CE	1
DE	0

In the next round, we don't need to generate itemsets involving AD, CD, CE, DE

TID	Items bought
1	A B E
2	B D
3	B C
4	A B D
5	A C
6	B C
7	A C
8	A B C E
9	A B C

minSupport count ≥ 2

3-itemsets	
Itemset	Count
ABC	2
ABE	2

Don't generate ABCE
because we already know
that CE is infrequent

Without support-based pruning, we have to
check the support of $2^5 = 32$ candidates
(see Slide 10)

But with Apriori, we check the support of
only $5 + 10 + 2 = 17$ candidates

Factors affecting complexity

⊕ minSupport threshold

- ▶ Low minSupport results in more itemsets passing threshold, and thus long frequent itemsets (more rounds of frequent itemset generation)
- ▶ Hence, larger search space

⊕ Recall runtime complexity $O(MNw)$. Although some candidates are pruned out, the following parameters still affect runtime

- ▶ Dimensionality (number of items, M) $\rightarrow$ number of rounds to run frequent itemset generation
- ▶ Size of database (number of transactions, N) and average length of transactions (w) $\rightarrow$ when scanning DB to count the support of each candidate itemset

FP-Growth

1. FP-tree construction

- 1.1 Create header table containing individual item in decreasing order of support count. Remove items that fail minSupport. Also rearrange items in each transaction according to this order
- 1.2 Read each transaction. Create frequent pattern tree (FP-tree) and update header table

2. FP-growth (or FP-tree mining)

For each item in header table, in reverse order {

- 2.1 Suffix = item in focus
- 2.2 Given all its prefix paths, create conditional FP-trees (i.e. FP-tree for that suffix)
- 2.3 For each path in conditional FP-tree + suffix, create frequent patterns from all combinations of items

}

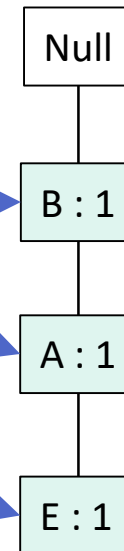
Example : FP-tree construction

Item	Count
B	7
A	6
C	6
D	2
E	2

TID	Transaction
1	B A E
2	B D
3	B C
4	B A D
5	A C
6	B C
7	A C
8	B A C E
9	B A C

Header Table		
Item	Count	Ptr
B	1	
A	1	
C		
D		
E	1	

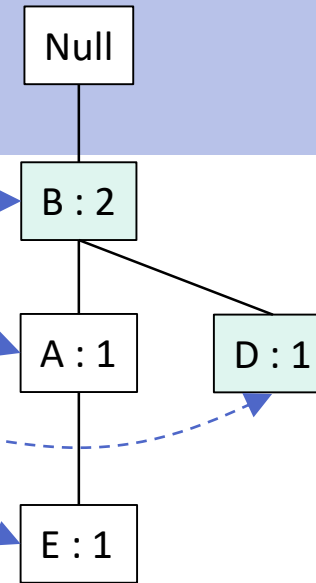
Pointers allow quick access to nodes in FP-tree



Items in each transaction are sorted by support

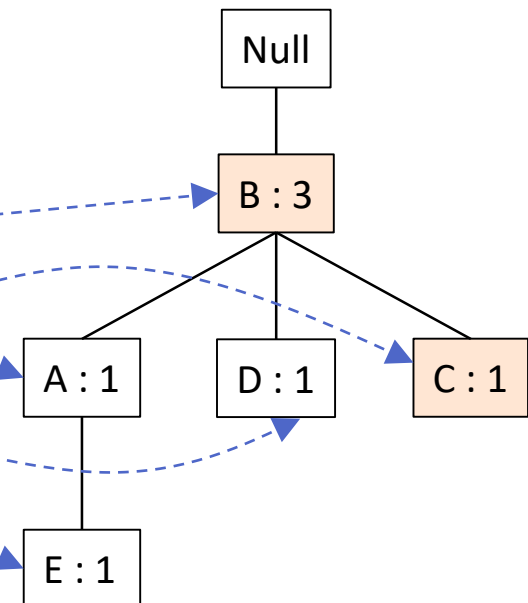
TID	Transaction
2	B D
3	B C
4	B A D
5	A C
6	B C
7	A C
8	B A C E
9	B A C

Header Table		
Item	Count	Ptr
B	2	
A	1	
C		
D	1	
E	1	



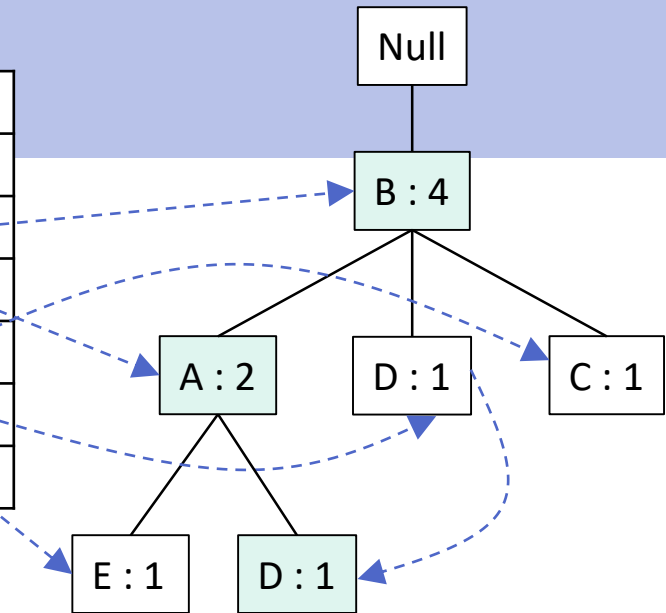
Update count if prefix nodes are common

Header Table		
Item	Count	Ptr
B	3	
A	1	
C	1	
D	1	
E	1	

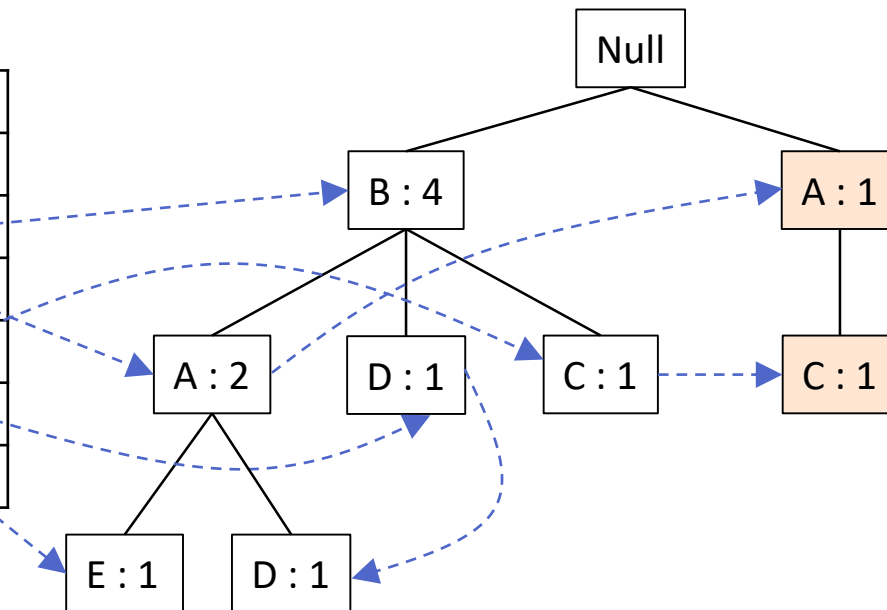


TID	Transaction
4	B A D
5	A C
6	B C
7	A C
8	B A C E
9	B A C

Header Table		
Item	Count	Ptr
B	4	
A	2	
C	1	
D	2	
E	1	

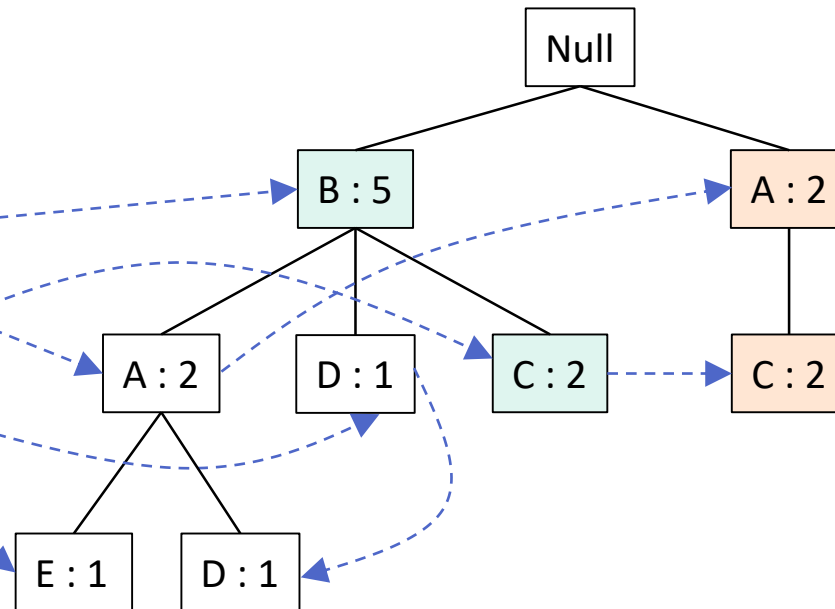


Header Table		
Item	Count	Ptr
B	4	
A	3	
C	2	
D	2	
E	1	



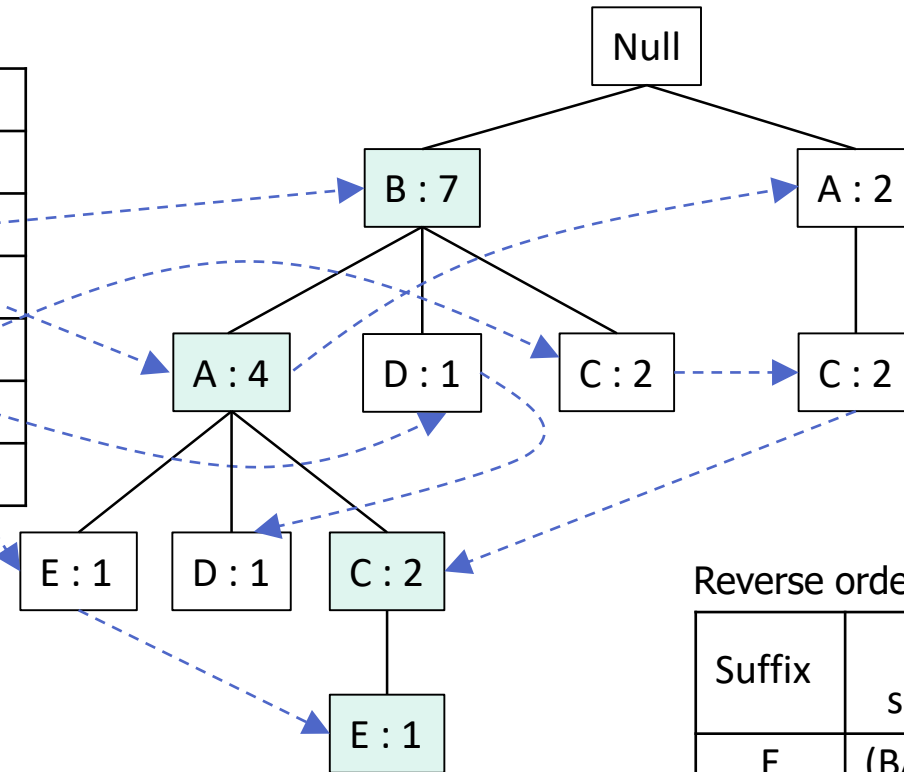
TID	Transaction
6	B C
7	A C
8	B A C E
9	B A C

Header Table		
Item	Count	Ptr
B	5	
A	4	
C	4	
D	2	
E	1	



TID	Transaction
8	B A C E
9	B A C

Header Table		
Item	Count	Ptr
B	7	
A	6	
C	6	
D	2	
E	2	

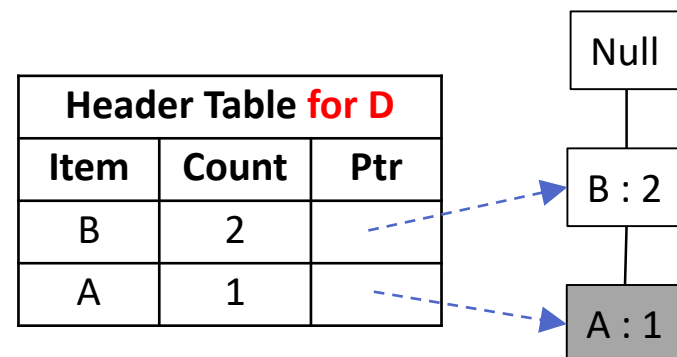
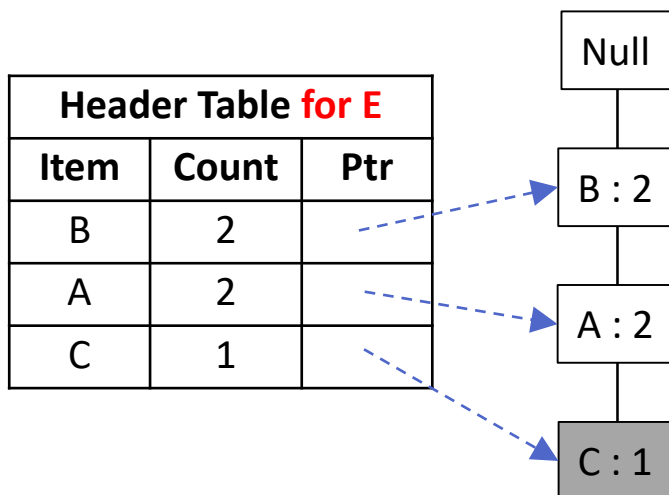


Reverse order of header table

Suffix	Prefix paths with support count of suffix
E	(BA : 1), (BAC : 1)
D	(B : 1), (BA : 1)
C	(B : 2), (A : 2), (BA : 2)
A	(B : 4)
B	Null

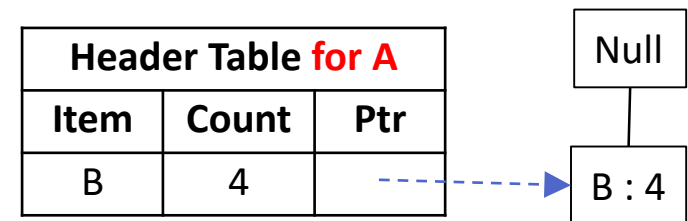
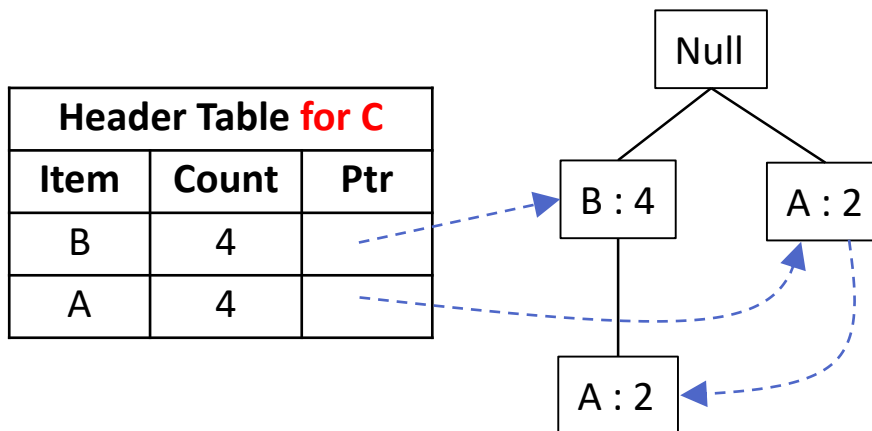
FP-growth

Suffix	Prefix paths with support count of suffix	Conditional FP-tree + suffix	Frequent Patterns
E	(BA : 1), (BAC : 1)	(B : 2, A : 2) + E	(BE : 2), (AE : 2), (BAE : 2)
D	(B : 1), (BA : 1)	(B : 2) + D	(BD : 2)



- Create conditional FP-tree in the same way as creating full FP-tree (prefix paths are treated as transactions). Prune out nodes that fail minSupport
- From each set of conditional FP-tree + suffix → generate frequent patterns from all item combinations where $support(FP) = \min\{\text{support of prefix items}\}$

Suffix	Prefix paths with support count of suffix	Conditional FP-tree + suffix	Frequent Patterns
C	(B : 2), (A : 2), (BA : 2)	(B : 4, A : 2) + C (A : 2) + C	(BC : 4), (AC : 2), (BAC : 2) (AC : 2) Identical patterns are merged (BC : 4), (AC : 4), (BAC : 2)
A	(B : 4)	(B : 4) + A	(BA : 4)



Phase 2 : Generating Association Rules

- ⊕ For each frequent itemset (or frequent pattern) FP
 - ▶ Permute items to get association rule. For k-itemset there are $2^k - 2$ possible permutations, excluding "FP $\rightarrow \emptyset$ " and " $\emptyset \rightarrow$ FP"
 - ▶ Keep only rules that pass minConfidence
- ⊕ Possible rules for frequent itemset BCE

$$conf(rule) = \frac{support(rule)}{support(premise)}$$

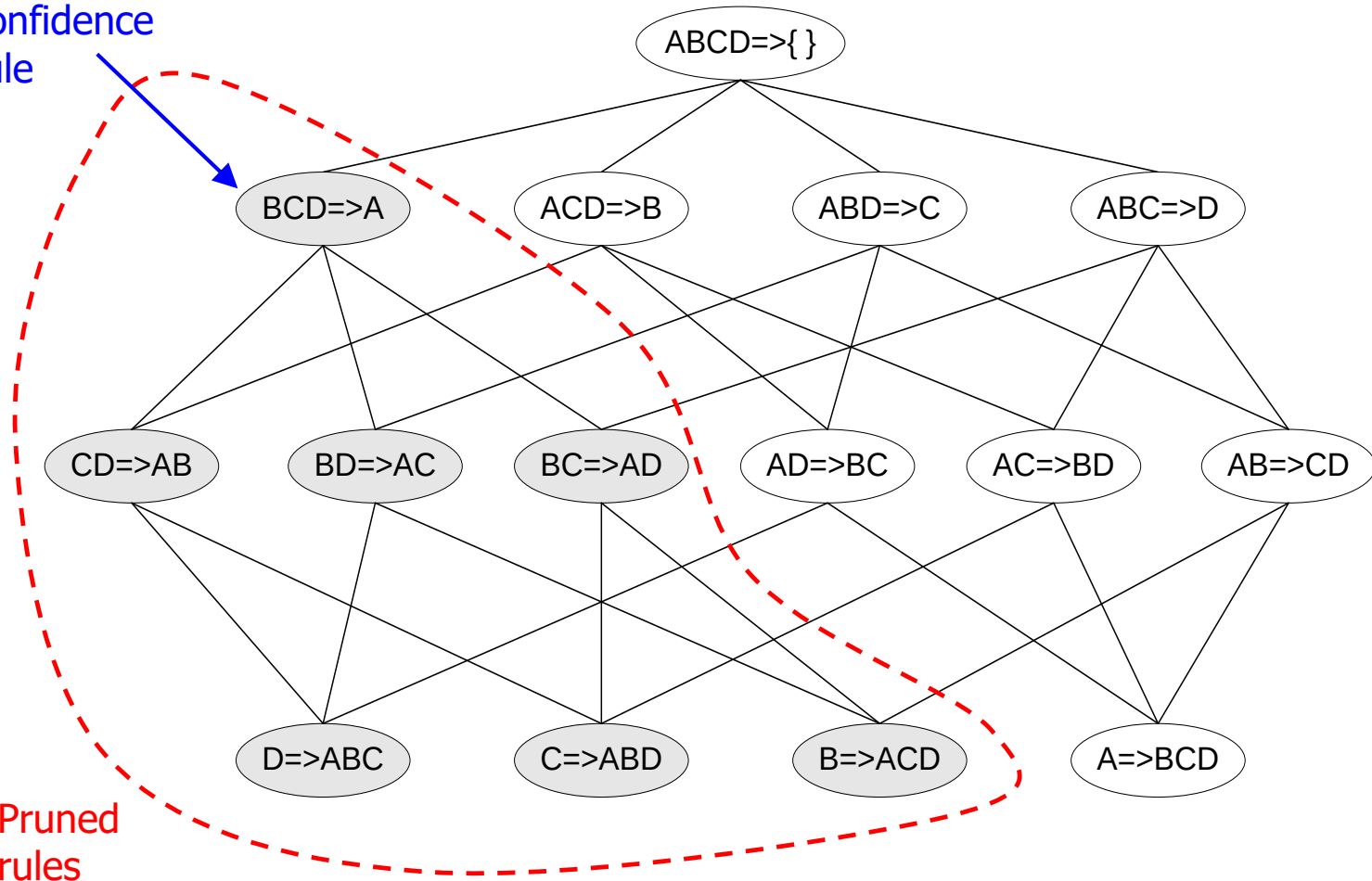
 - ▶ BC $\rightarrow$ E, BE $\rightarrow$ C, CE $\rightarrow$ B
 - ▶ E $\rightarrow$ BC, C $\rightarrow$ BE, B $\rightarrow$ CE
 - ▶ But they have different confidence. For example

BC $\rightarrow$ E, confidence = 90%	If we know a customer has bought B and C, we can be highly confidence that she'll buy E
E $\rightarrow$ BC, confidence = 40%	If we know a customer has bought E, we can be only moderately confidence that she'll buy B and C

Confidence & Anti-Monotone Property

- ✦ Confidence has no anti-monotone property w.r.t size of premise (on LHS)
 - ▶ $\text{Conf}(\mathbf{B} \rightarrow \mathbf{E})$ can be higher or lower than $\text{conf}(\mathbf{BC} \rightarrow \mathbf{E})$; $\mathbf{B} \subseteq \mathbf{BC}$
- ✦ But for rules generated from the same itemset, anti-monotone property exists w.r.t size of premise (on LHS) and conclusion (on RHS)
 - ▶ Given frequent itemset ABCD
 - ▶ $\text{Conf}(\text{rule}) = \text{support}(\text{ABCD}) / \text{support}(\text{premise})$
 - ▶ $\text{Conf}(\mathbf{ABC} \rightarrow \mathbf{D}) \geq \text{conf}(\mathbf{AB} \rightarrow \mathbf{CD}) \geq \text{conf}(\mathbf{A} \rightarrow \mathbf{BCD})$; $\mathbf{A} \subseteq \mathbf{AB} \subseteq \mathbf{ABC}$
 - ▶ Because $\text{support}(\text{ABCD})$ is the same for all rules and by anti-monotone property of support, $\text{support}(\text{ABC}) \leq \text{support}(\text{AB}) \leq \text{support}(\text{A})$
 - 3 evidences (ABC) should give more confidence than just 1 (A)
- ✦ Use confidence-based pruning to prune out some candidate rules
 - ▶ If $\mathbf{ABC} \rightarrow \mathbf{D}$ fails minConfidence, we don't need to check $\mathbf{A} \rightarrow \mathbf{BCD}$

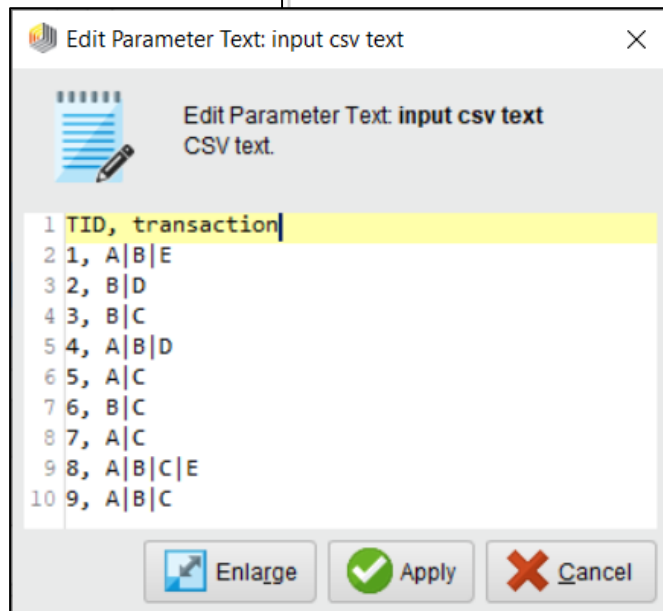
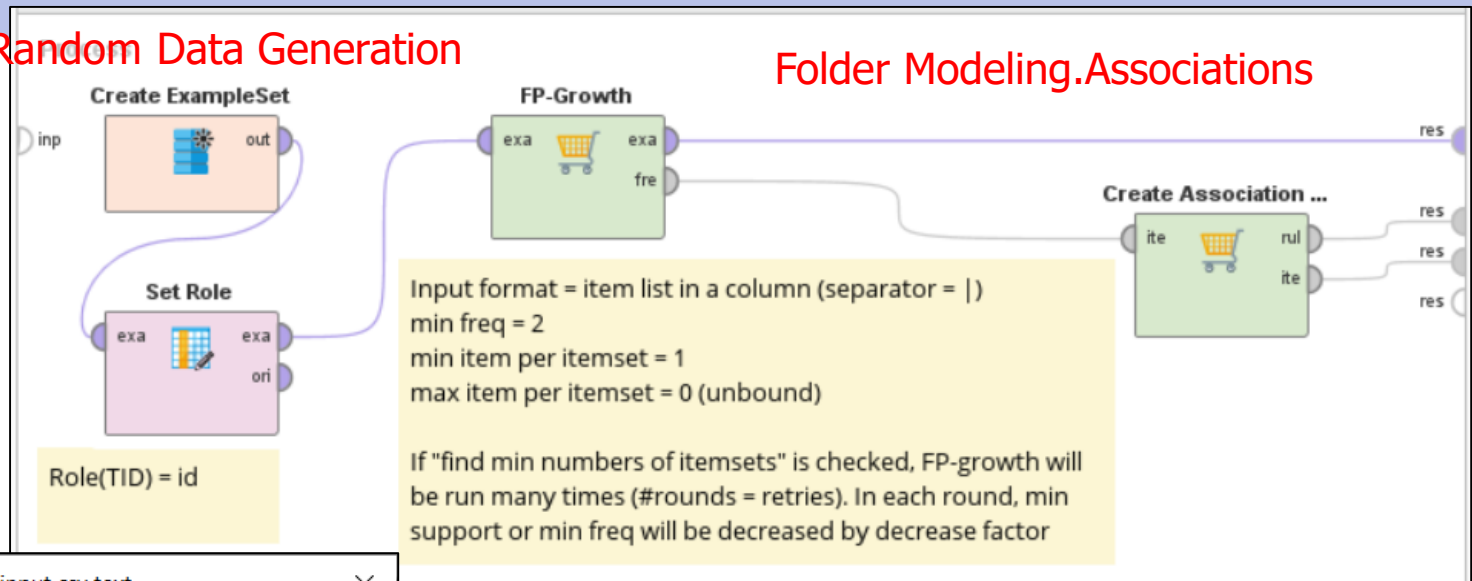
Low
confidence
rule



Example (1) : transaction input

Folder Utility.Random Data Generation

Folder Modeling.Associations



TID must be set to ID role, so that it is excluded from itemset generation

Min item per itemset = 1 to include 1-itemset. Otherwise, RapidMinder will give wrong metric calculation (e.g. confidence) when generating association rules

No. of Sets: 13
Total Max. Size: 3

Min. Size:
Max. Size:
Contains Item:

FP filtering

Size	Support	Item 1	Item 2	Item 3
1	0.778	B		
1	0.667	A		
1	0.667	C		
1	0.222	D		
1	0.222	E		
2	0.444	B	A	
2	0.444	B	C	
2	0.222	B	D	
2	0.222	B	E	
2	0.444	A	C	
2	0.222	A	E	
3	0.222	B	A	C
3	0.222	B	A	E

We need these
for metric calculation
(next slide)

$$\text{Conf}(B \rightarrow A) = \text{support}(B \rightarrow A) / \text{support}(B) \\ = 0.444 / 0.778 = 0.571$$

Show rules matching

all of these conclusions: ▼

B
A
C
E

Rule filtering

Min. Criterion:

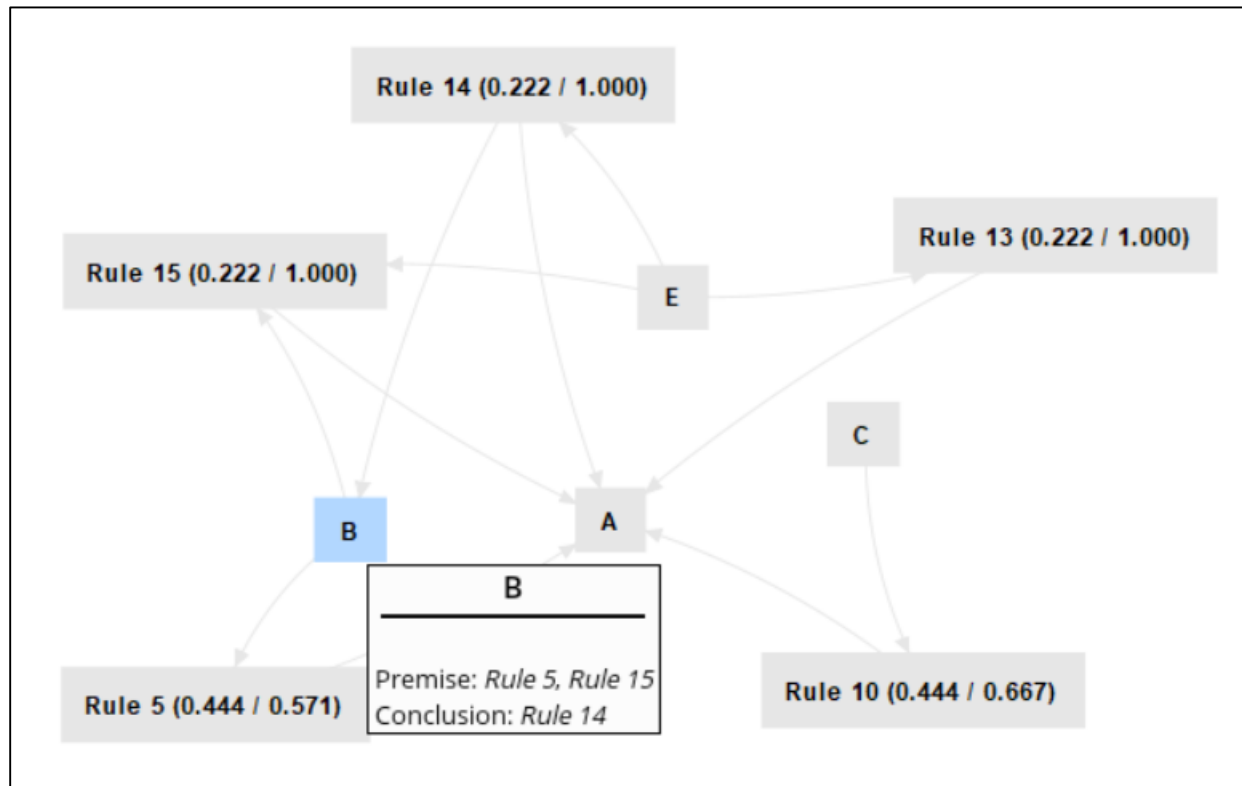
confidence ▼

Min. Criterion Value:

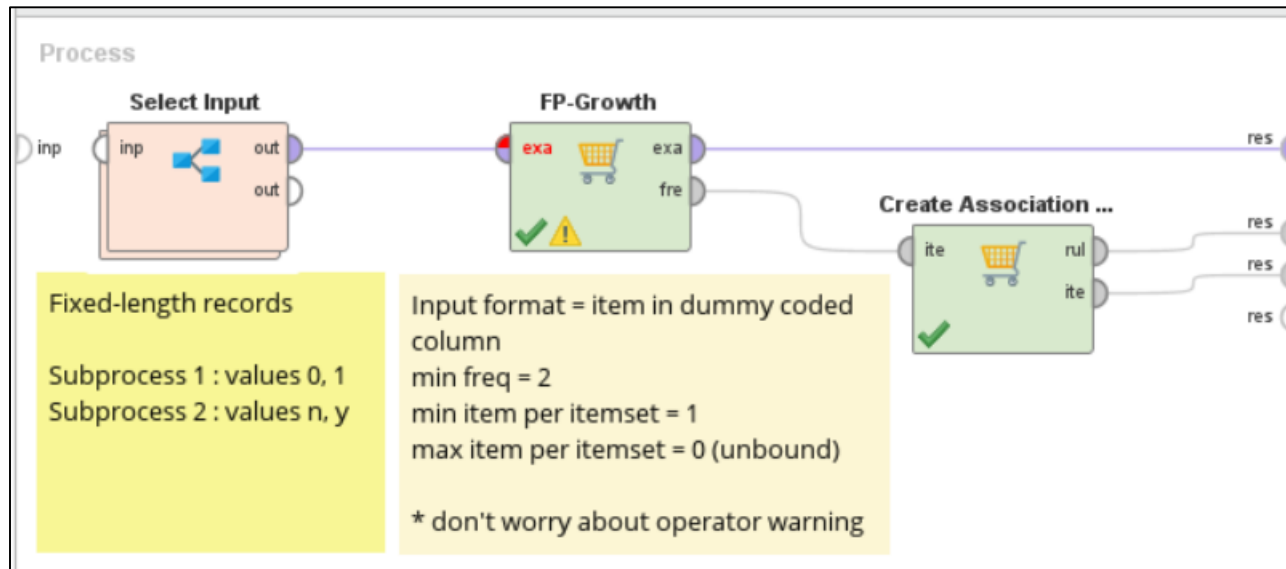


No.	Premises	Conclusion	Support	Confidence	Lift
5	B	A	0.444	0.571	0.857
6	B	C	0.444	0.571	0.857
7	A	B	0.444	0.667	0.857
8	C	B	0.444	0.667	0.857
9	A	C	0.444	0.667	1
10	C	A	0.444	0.667	1
11	D	B	0.222	1	1.286
12	E	B	0.222	1	1.286
13	E	A	0.222	1	1.500
14	E	B, A	0.222	1	2.250
15	B, E	A	0.222	1	1.500
16	A, E	B	0.222	1	1.286

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Example (2) : fixed-length record input

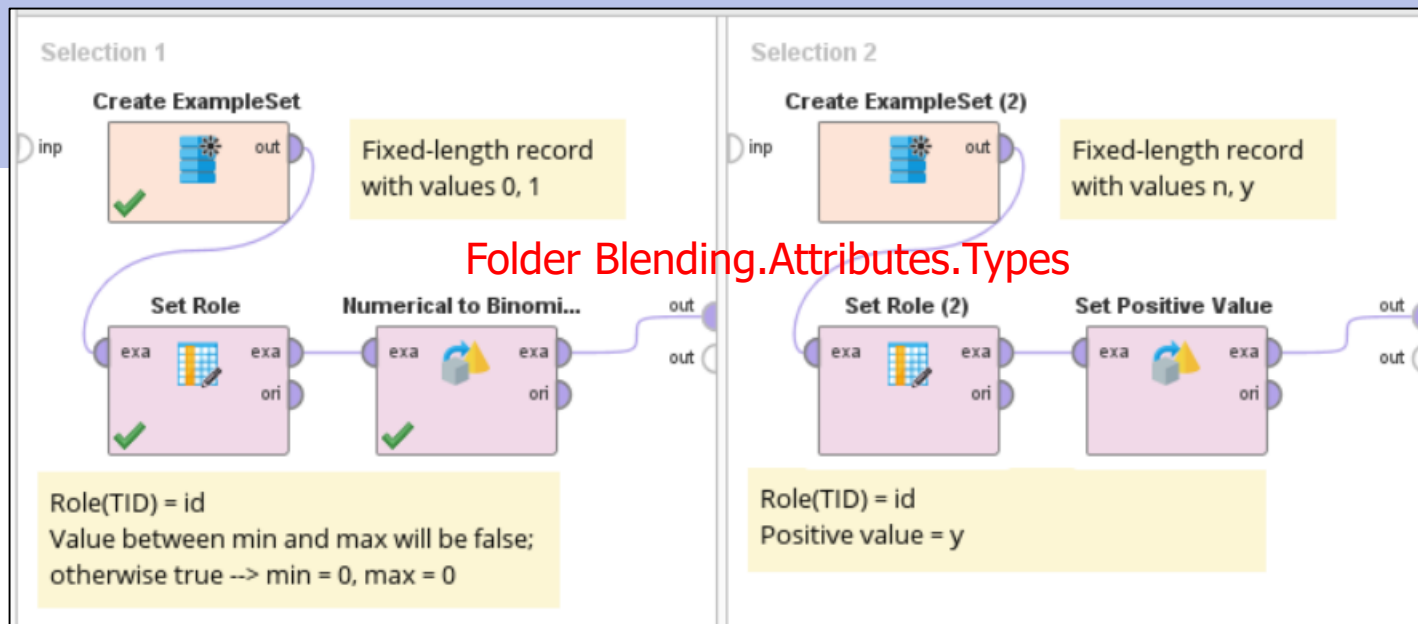


Item attributes must be binominal or nominal with only 2 values

Asymmetric attribute setup

True = presence, i.e. item is bought ; False = absence

In case that values $\neq \{ \text{false}, \text{true} \}$ $\rightarrow$ set which one is positive, i.e. to be treated as true



Edit Parameter Text: input csv text

Edit Parameter Text: input csv text
CSV text.

1	TID, A, B, C, D, E
2	1, 1, 1, 0, 0, 1
3	2, 0, 1, 0, 1, 0
4	3, 0, 1, 1, 0, 0
5	4, 1, 1, 0, 1, 0
6	5, 1, 0, 1, 0, 0
7	6, 0, 1, 1, 0, 0
8	7, 1, 0, 1, 0, 0
9	8, 1, 1, 1, 0, 1
10	9, 1, 1, 1, 0, 0

Enlarge Apply Cancel

Edit Parameter Text: input csv text

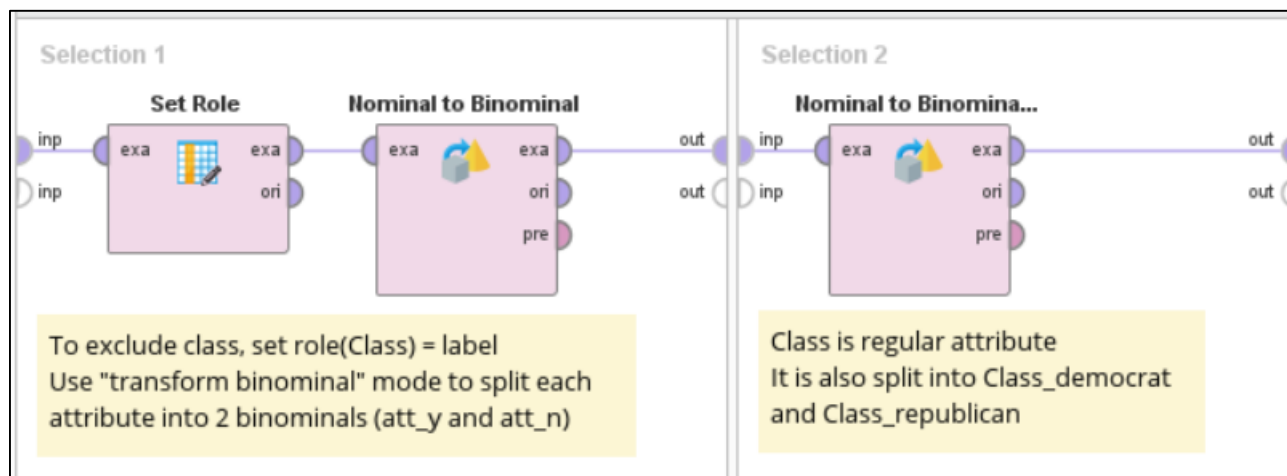
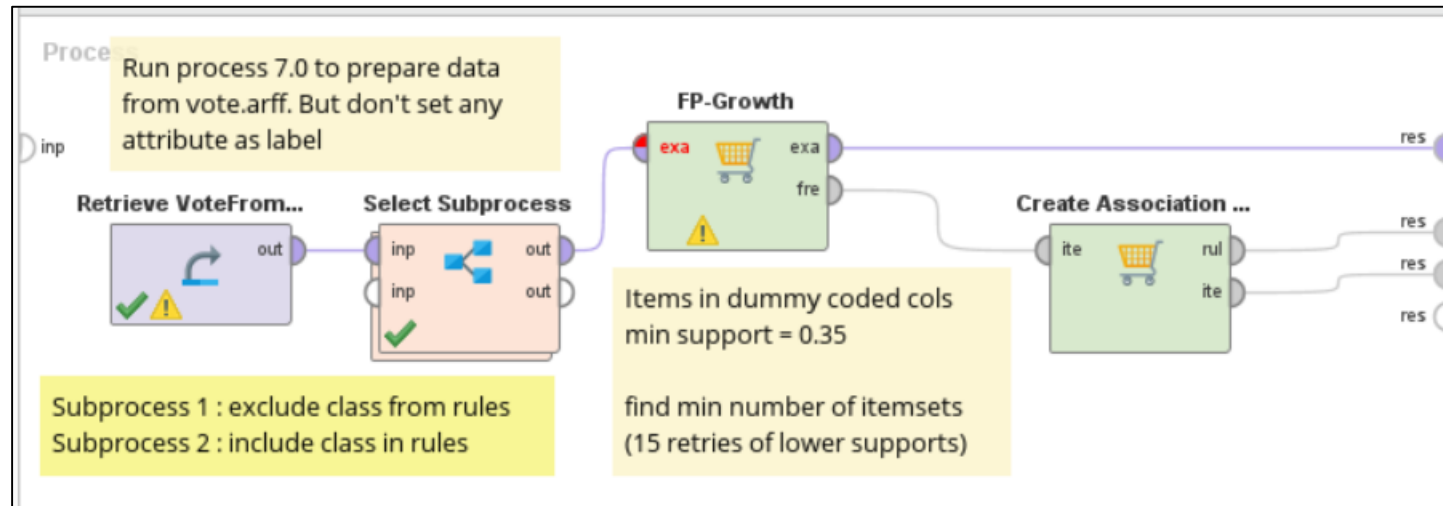
Edit Parameter Text: input csv text
CSV text.

1	TID, A, B, C, D, E
2	1, y, y, n, n, y
3	2, n, y, n, y, n
4	3, n, y, y, n, n
5	4, y, y, n, y, n
6	5, y, n, y, n, n
7	6, n, y, y, n, n
8	7, y, n, y, n, n
9	8, y, y, y, n, y
10	9, y, y, y, n, n

Enlarge Apply

Same results as
Example (1)

Example (3) : symmetric attributes



Subprocess 1 : excluding class (as special attribute)

After nominal to binominal transformation

Row No.	id	Class	handicapped-infants_n	handicapped-infants_y
1	1	republican	true	false
2	2	republican	true	false
3	3	democrat	false	false
4	4	democrat	true	false

Rules with both yes and no votes, but none of them contains Class

No.	Premises	Conclusion	Support	Confidence	Lift
65	aid-to-nicaraguan-contras_y, anti-satellite-test-ban_y	mx-missile_y	0.398	0.824	1.731
68	physician-fee-freeze_n, aid-to-nicaraguan-contras_y	mx-missile_y	0.400	0.825	1.733
77	physician-fee-freeze_n, anti-satellite-test-ban_y	mx-missile_y	0.375	0.827	1.739
85	adoption-of-the-budget-resolution_y, aid-to-nicaraguan-co...	mx-missile_y	0.356	0.829	1.742
89	anti-satellite-test-ban_y, education-spending_n	mx-missile_y	0.352	0.832	1.747
103	aid-to-nicaraguan-contras_y, education-spending_n	mx-missile_y	0.372	0.835	1.755

Subprocess 2 : including class (as regular attribute)

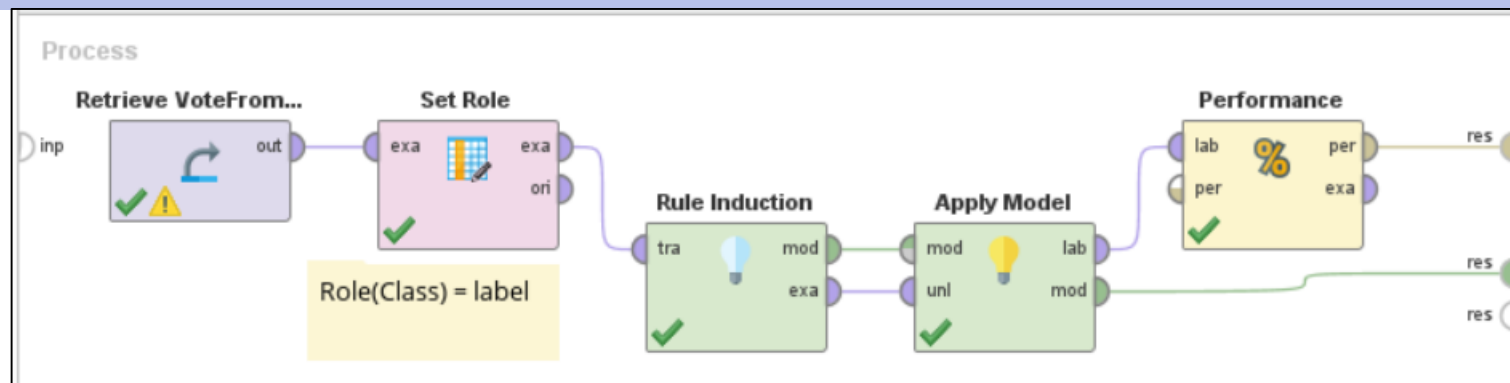
After nominal to binominal transformation

Row No.	id	Class_democrat	Class_republican	handicapped-infants_n	handicapped-infants_y
1	1	false	true	true	false
2	2	false	true	true	false
3	3	true	false	false	false
4	4	true	false	true	false

Vote patterns of Democrat, with both yes and no votes

No.	Premises	Conclusion	Support	Confidence	Lift
1448	physician-fee-freeze_n, anti-satellite-test-ban_y	Class_democrat	0.448	0.990	1.613
1449	physician-fee-freeze_n	Class_democrat	0.563	0.992	1.616
1451	physician-fee-freeze_n, aid-to-nicaraguan-contras_y, el-s...	Class_democrat	0.352	0.994	1.619
1452	physician-fee-freeze_n, anti-satellite-test-ban_y, el-salvad...	Class_democrat	0.352	0.994	1.619
1453	physician-fee-freeze_n, duty-free-exports_y	Class_democrat	0.354	0.994	1.619
1454	physician-fee-freeze_n, anti-satellite-test-ban_y, educatio...	Class_democrat	0.354	0.994	1.619

Example (4) : comparing with rule induction



RuleModel

```

if physician-fee-freeze = n then democrat (2 / 245)
else republican (149 / 19)

```

correct: 394 out of 415 training examples.

Accuracy = 94.48%

Association rules in Example (3) **describe** vote patterns of Democrats and Republicans on various issues, i.e. how these issues are associated

Classification rules in this example **predict** Democrat or Republican from physician-fee-freeze vote which is the most predictive attribute

Evaluating Association Rules

- ⊕ We don't measure correct/incorrect prediction as in classification
- ⊕ Subjective evaluation
 - ▶ Rules revealing obvious fact are subjectively uninteresting, e.g.
 - {butter} → {bread} is uninteresting
 - {diaper} → {beer} is unexpected & suggests new cross-selling opportunity
 - ▶ Require prior knowledge about the domain
 - ▶ Visualization is useful for constructing subjective argument
- ⊕ Objective evaluation, i.e. based on some measures
 - ▶ Support
 - ▶ Confidence
 - ▶ Lift

Subjective Interestingness

⊕ A pattern is subjectively interesting if

- ▶ It contradicts user's expectation
- ▶ It is actionable

+ Pattern expected to be frequent

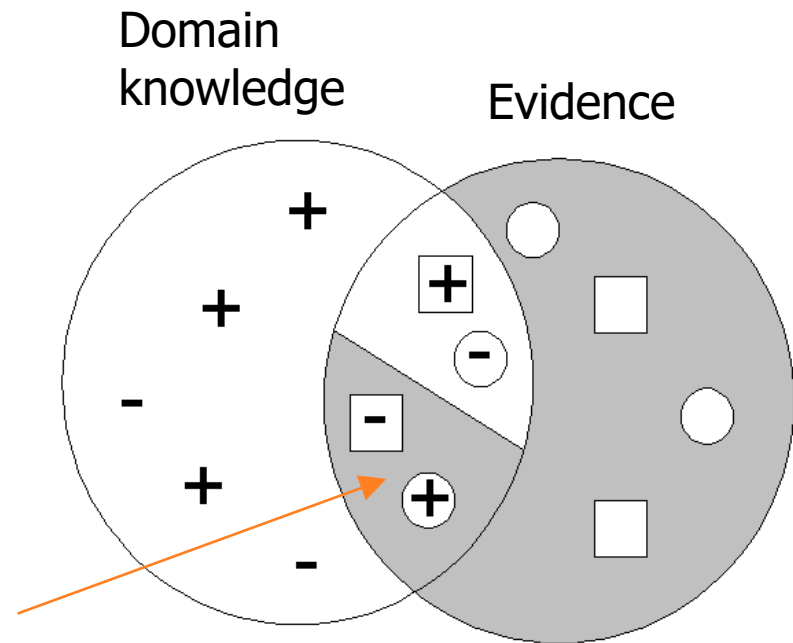
- Pattern expected to be infrequent

□ Pattern found to be frequent

○ Pattern found to be infrequent

⊕ Expected patterns

⊖ Unexpected (interesting) patterns



Objective Evaluation

✦ Construct a contingency table for frequency counts

✦ For example, rule "tea $\rightarrow$ coffee"

	coffee	~coffee	
tea	15	5	20
~tea	75	5	80
	90	10	100

Support = $P(\text{tea} \cap \text{coffee}) = 0.15$

Confidence = $P(\text{coffee} \mid \text{tea})$
 $= 15/20 = \underline{0.75}$

Confidence is high but misleading

✦ "Coffee regardless of tea" is very high, i.e. $P(\text{coffee}) = 0.90$

✦ Consider rule " $\sim$ tea $\rightarrow$ coffee"

▶ Confidence = $P(\text{coffee} \mid \sim\text{tea}) = 75/80 = \underline{0.9375}$

▶ This is actually **negative association** between tea and coffee

Lift

⊕ Independent items vs. confidence

- ▶ From confidence($A \rightarrow B$) = $P(B | A) = P(A \cap B) / P(A)$
- ▶ If A and B are independent, then $P(A \cap B) = P(A) \times P(B)$
 - Therefore, confidence($A \rightarrow B$) = $\frac{P(A) \times P(B)}{P(A)} = P(B)$
 - A and B are not associated, despite possibly high confidence

⊕ Lift or interest : take out independence factor

- ▶ $\text{Lift}(A \rightarrow B) = \text{confidence}(A \rightarrow B) / \text{support}(B)$
 $= P(A \cap B) / (P(A) \times P(B))$
- ▶ $\text{Lift}(A \rightarrow B) < 1$ implies negative association
- ▶ $\text{Lift}(A \rightarrow B) = 1$ no association
- ▶ $\text{Lift}(A \rightarrow B) > 1$ positive association

	coffee	~coffee	
tea	15	5	20
~tea	75	5	80
	90	10	100

- $\text{Lift}(\text{tea} \rightarrow \text{coffee}) = P(\text{tea} \cap \text{coffee}) / (P(\text{tea}) * P(\text{coffee}))$
 $= 0.15 / (0.2 * 0.9) = \underline{0.83}$
- $\text{Lift}(\sim\text{tea} \rightarrow \text{coffee}) = P(\sim\text{tea} \cap \text{coffee}) / (P(\sim\text{tea}) * P(\text{coffee}))$
 $= 0.75 / (0.8 * 0.9) = \underline{1.04}$
- Rule " $\sim\text{tea} \rightarrow \text{coffee}$ " should be used to describe their association instead (customers only buy either of them, not both)

Limitation of Lift

$R \rightarrow S$

	S	$\sim S$	
R	75	5	80
$\sim R$	10	10	20
	85	15	100

$X \rightarrow Y$

	Y	$\sim Y$	
X	10	5	15
$\sim X$	10	75	85
	20	80	100

- ⊕ $\text{Lift}(R \rightarrow S) = P(R \cap S) / (P(R) * P(S)) = 0.75 / (0.8 * 0.85) = \underline{1.1}$
- ⊕ $\text{Lift}(X \rightarrow Y) = P(X \cap Y) / (P(X) * P(Y)) = 0.10 / (0.15 * 0.2) = \underline{3.3}$
- ⊕ $X \rightarrow Y$ has higher lift, but they hardly co-exist
- ⊕ $\text{Confidence}(R \rightarrow S) = \underline{0.9375} > \text{confidence}(X \rightarrow Y) = \underline{0.67}$
- ⊕ Lift is used to ensure positive association (we only want it to be >1). Then, use support & confidence to measure coverage & reliability of the rule

Rules in Opposite Directions

$R \rightarrow S$

	S	$\sim S$	
R	75	5	80
$\sim R$	10	10	20
	85	15	100

⌘ Support measures the coverage of frequent pattern $RS \rightarrow$ same support for both directions

⌘ Lift measures the association between premise & conclusion $\rightarrow$ same lift for both directions

$R \rightarrow S$	$S \rightarrow R$
Support = $P(R \cap S)$	Support = $P(S \cap R)$
Confidence = $P(R \cap S) / P(R)$	Confidence = $P(S \cap R) / P(S)$
Lift = $P(R \cap S) / (P(R) * P(S))$	Lift = $P(S \cap R) / (P(S) * P(R))$

⌘ Confidence measures the reliability of conclusion given premise $\rightarrow$ the values can be different, depending on premises (i.e. divisors)

Simson's Paradox

⊕ Scenario 1 : association between TV and exercise machine sales

	XM	~XM	
TV	99	81	180
~TV	54	66	120
	153	147	300

- ▶ $\text{Conf}(\text{TV} \rightarrow \text{XM}) = 99/180 = 0.55$
- ▶ $\text{Lift}(\text{TV} \rightarrow \text{XM}) = 0.55 / (153/300) = 1.07$ slightly positive

⊕ Scenario 2 : customers are divided into 2 groups

		XM	~XM		
College Students	TV	1	9	10	
	~TV	4	30	34	Total = 44
Working Adults	TV	98	72	170	
	~TV	50	36	86	Total = 256

► Students

- $\text{Conf}(\text{TV} \rightarrow \text{XM}) = 1/10 = 0.1$
- $\text{Lift}(\text{TV} \rightarrow \text{XM}) = 0.1 / (5/44) = 0.88$ slightly negative

► Working adults

- $\text{Conf}(\text{TV} \rightarrow \text{XM}) = 98/170 = 0.5765$
- $\text{Lift}(\text{TV} \rightarrow \text{XM}) = 0.5765 / (148/256) = 0.997$ slightly negative

If we use conf = 0.58, lift will be 1.003

- ⊕ Slight change in frequency distribution leads to different conclusion
- ⊕ In practice, we prefer scenario 2
- ⊕ So we should understand domain background in order to identify and handle potentially **confound factors**, e.g.
 - ▶ TV and exercise machine example confound = customer type
 - ▶ Supermarket sale analysis confound = store location
 - ▶ Medical analysis confound = age, gender
 - ▶ Running association analysis on each group may give better results

Summary

- ⊕ Association rule describes what conclusion is likely to occur given premise or evidence from data
 - ▶ It is mainly descriptive data mining task
 - ▶ We can also use association methods to generate classification rules. This approach is called associative classification, but it is not covered in this course
- ⊕ Attributes are usually asymmetric binary attributes → focusing on only positive value or presence
- ⊕ FP-growth is generally faster than Apriori, especially when many transactions (or records) share common items

- ⊕ Objective = strong positive association rules
 - ▶ Strong rule has high support and high confidence
 - ▶ Positive rule ($\text{lift} > 1$) indicates that premise and conclusion occur together → e.g. customers buy both items
 - ▶ Negative association ($\text{lift} < 1$) can be interpreted as negating ($\sim$) one side of the rule → e.g. customers buy either item
- ⊕ It is common to get a very large number of rules, with different premise & conclusion patterns
 - ▶ Visualization and interactive output (as in RapidMiner) are useful for searching, filtering, interpreting patterns